

Geminals

Every geminal $|g\rangle \in \mathcal{H}^2$ defines a symplectic form on $\mathcal{H}^1$ given by

$$g(\varphi_1, \varphi_2) = \langle g | \varphi_1 \wedge \varphi_2 \rangle = -g(\varphi_2, \varphi_1); |\varphi_1\rangle, |\varphi_2\rangle \in \mathcal{H}^1 \quad \#$$

can be expressed in canonical form as

$$|g\rangle = \sum_{1 \leq j \leq s} c_j |\psi_j \psi_{-j}\rangle = \sum_{1 \leq j \leq s} n_j^{\frac{1}{2}} |\psi_j \psi_{-j}\rangle = \sum_{1 \leq j \leq s} \left| c_j^{\frac{1}{2}} \psi_j c_j^{\frac{1}{2}} \psi_{-j} \right\rangle = \sum_{1 \leq j \leq s} |\eta_j \eta_{-j}\rangle, \quad \#$$

where $\{|\psi_j\rangle | -s \leq j \leq s\}$ is a complete orthonormal basis for $\mathcal{H}^1$ and $c_j \geq 0, 1 \leq j \leq s$. (The pairs GSO's $|\psi_j\rangle, |\psi_{-j}\rangle$ are not unique as $|\psi_j \psi_{-j}\rangle$ are invariant under $SU(2)$).

Antisymmetric Operators and Geminals

The space of antisymmetric Hilbert-Schmidt operators, $\mathcal{B}_{2a}(\mathcal{H}^1)$ can be defined as

$$\begin{aligned} \mathcal{B}_{2a}(\mathcal{H}^1) &= \left\{ \sum_{1 \leq j, k \leq r} g_{\psi j k} |\psi_j\rangle \langle \psi_k| \left| g_{jk} = -g_{kj}, 1 \leq j, k \leq r, g_{jk} \in \mathbb{C}, \right. \right. \\ &\quad \left. \{|\psi_j\rangle | 1 \leq j \leq r\} \text{ a fixed conb for } \mathcal{H}^1 \right\} \\ &= \left\{ T \mathbf{g}_\psi T^t \left| T \in \frac{GL(2s, \mathbb{C})_{\mathcal{H}^1}}{Sp(2s, \mathbb{C})_{\mathcal{H}^1}} \right. \right\}. \end{aligned} \quad \#$$

Change of Basis

A change of basis in $\mathcal{H}^1$ produces the following effect in $\mathcal{B}_{2a}(\mathcal{H}^1)$

$$\begin{aligned} \sum_{1 \leq j, k \leq r} g_{\chi j k} |\chi_j\rangle \langle \chi_k| &= \sum_{1 \leq j, k \leq r} g_{\chi j k} |S\psi_j\rangle \langle S\psi_k| = \\ &= \sum_{1 \leq j, k, l, m \leq r} S_{jl} \bar{S}_{km} g_{\chi j k} |\psi_j\rangle \langle \psi_k| = \sum_{1 \leq j, k \leq r} g_{\psi j k} |\psi_j\rangle \langle \psi_k| \end{aligned} \quad \#$$

thus

$$\mathbf{g}_\chi = \mathbb{S}^t \mathbf{g}_\psi \bar{\mathbb{S}} \quad \#$$

and

$$\mathbf{g}_\chi = -\mathbf{g}_\chi^t \text{ iff } \mathbb{S} = \bar{\mathbb{S}}. \quad \#$$

This should be contrasted to the effect produced in $\mathcal{H}^2$ by a change of basis in $\mathcal{H}^1$ i.e

$$\mathbf{g}_\chi = \mathbb{S}^t \mathbf{g}_\psi \mathbb{S}. \quad \#$$

Normalization

$$\langle T^t g T | T^t g T \rangle = 1 = \text{Tr} \{ T^\dagger \mathbf{g}^\dagger \bar{T} T^t \mathbf{g} T \}, \quad \#$$

so in particular if $T \in U(2s)$ the norm is conserved.

Mapping between the Spaces

Consider the map M_a between $\mathcal{H}^2$ and the space of antisymmetric Hilbert-Schmidt operators, $\mathcal{B}_{2a}(\mathcal{H}^1)$, i.e.

$$M_a : \mathcal{H}^2 \rightarrow \mathcal{B}_{2a}(\mathcal{H}^1) \quad \#$$

given by

$$M_a(|g\rangle) = M_a\left(\sum_{1 \leq j \leq s} c_j |\psi_j \psi_{-j}\rangle\right) = \sum_{1 \leq j \leq s} c_j \{|\psi_j\rangle\langle\psi_{-j}| - |\psi_{-j}\rangle\langle\psi_j|\}. \quad \#$$

One can then observe that

$$\begin{aligned} M_a\left(\sum_{1 \leq j \leq s} c_j |\psi_j \psi_{-j}\rangle\right) &= M_a\left(\sum_{1 \leq j \leq s} c_j |e^{i\theta_j} \psi_j e^{-i\theta_j} \psi_{-j}\rangle\right) \\ &= \sum_{1 \leq j \leq s} c_j \{e^{2i\theta_j} |\psi_j\rangle\langle\psi_{-j}| - e^{-2i\theta_j} |\psi_{-j}\rangle\langle\psi_j|\} \end{aligned} \quad \#$$

thus M_a is multivalued. This can also be expressed using

$$\begin{aligned} |\chi_j(\theta_j)\rangle &= |e^{i\theta_j} \psi_j\rangle \\ |\chi_{-j}(\theta_j)\rangle &= |e^{-i\theta_j} \psi_{-j}\rangle \end{aligned} \quad \#$$

producing the antisymmetric operator

$$\sum_{1 \leq j \leq s} c_j \{|\chi_j(\theta_{j1})\rangle\langle\chi_{-j}(\theta_{j1})| - |\chi_{-j}(\theta_{j1})\rangle\langle\chi_j(\theta_{j1})|\}. \quad \#$$

This transformation corresponds to the one parameter subgroup $U(1)$ of $SU(2)$ generated by

$$u(\theta) = e^{i\theta_j(|\psi_j\rangle\langle\psi_j| - |\psi_{-j}\rangle\langle\psi_{-j}|)}. \quad \#$$

If we consider the "inverse" map M_b defined as

$$M_b\left(\sum_{1 \leq j \leq s} c_j \{|\psi_j\rangle\langle\psi_{-j}| - |\psi_{-j}\rangle\langle\psi_j|\}\right) = \sum_{1 \leq j \leq s} c_j |\psi_j \psi_{-j}\rangle = |g\rangle \quad \#$$

then

$$\begin{aligned} M_b\left(e^{i\theta_j(|\psi_j\rangle\langle\psi_j| - |\psi_{-j}\rangle\langle\psi_{-j}|)} \sum_{1 \leq j \leq s} c_j \{|\psi_j\rangle\langle\psi_{-j}| - |\psi_{-j}\rangle\langle\psi_j|\}\right) \\ = M_b\left(\sum_{1 \leq j \leq s} c_j \{e^{2i\theta_j} |\psi_j\rangle\langle\psi_{-j}| - e^{-2i\theta_j} |\psi_{-j}\rangle\langle\psi_j|\}\right) \\ = \sum_{1 \leq j \leq s} c_j |\psi_j \psi_{-j}\rangle. \end{aligned} \quad \#$$

In general

$$M_b(u(\theta)A) = M_b(A), \quad A \in \mathcal{B}_{2a}(\mathcal{H}^1), u(\theta) \in U(1) \quad \#$$

and M_b is singular (it is surjective but not injective). This gives

$$M_b\left(\sum_{1 \leq j \leq s} c_j \{(e^{2i\theta_{1j}} - e^{2i\theta_{2j}})|\psi_j\rangle\langle\psi_{-j}| - (e^{-2i\theta_{1j}} - e^{-2i\theta_{2j}})|\psi_{-j}\rangle\langle\psi_j|\}\right) = 0. \quad \#$$

Clearly M_b is an isometry as

$$\langle M_b(A) | M_b(A) \rangle = \text{Tr}\{A^\dagger A\}. \quad \#$$

One should note that if $|g\rangle$ is non degenerate the maximal compact invariance group is the special unitary group $SU(2)^s$, while the maximal non compact one is $SL(2, \mathbb{C})^s$ i.e.

$$\sum_{1 \leq j \leq s} c_j |u\psi_j u\psi_{-j}\rangle = \sum_{1 \leq j \leq s} c_j |\psi_j \psi_{-j}\rangle, u \in SU(2)^s, \quad \#$$

$$\sum_{1 \leq j \leq s} c_j |v\psi_j v\psi_{-j}\rangle = \sum_{1 \leq j \leq s} c_j |\psi_j \psi_{-j}\rangle, v \in SL(2, \mathbb{C})^s, \quad \#$$

while if the degeneracy is twofold then the maximal compact invariance group is $SU(2)^{s-2} \times USp(4)$, while the non compact one is $SL(2, \mathbb{C})^{s-2} \times Sp(4, \mathbb{C})$ etc until one has an s -fold degeneracy then the maximal invariance groups $USp(2s)$ and $Sp(2s, \mathbb{C})$.

Transpose and Complex Conjugation

The transpose operation

$$\tau : \mathcal{B}(\mathcal{H}^1) \rightarrow \mathcal{B}(\mathcal{H}^1) \quad \#$$

$$A^t = \tau(A), \quad A \in \mathcal{B}(\mathcal{H}^1) \quad \#$$

is defined in terms of a complex conjugation map, K , that is an involutive antilinear map

$$K : \mathcal{H}^1 \rightarrow \mathcal{H}^1, \quad \#$$

$$K(v) = |v^*\rangle, \quad \#$$

that is associated with an equivalence of bases of $\mathcal{H}^1$. The matrix representation of an element of $\mathcal{H}^1$ and its K – complex conjugate are

$$|v\rangle = (|\lambda_1\rangle, \dots, |\lambda_s\rangle, |\lambda_{-1}\rangle, \dots, |\lambda_{-s}\rangle)\mathbf{v}$$

$$|v^*\rangle = (|\lambda_1\rangle, \dots, |\lambda_s\rangle, |\lambda_{-1}\rangle, \dots, |\lambda_{-s}\rangle)\bar{\mathbf{v}}. \quad \#$$

If

$$K|v\rangle = |v\rangle \quad \#$$

then $\mathbf{v} \in \mathbb{R}$. The complex conjugation map, K , also defines a non degenerate symmetric bilinear form on $\mathcal{H}^1$ given by

$$(\varphi|\chi) = \langle K\varphi|\chi\rangle \quad |\varphi\rangle, |\chi\rangle \in \mathcal{H}^1. \quad \#$$

The change of basis

$$S(|\lambda_1\rangle, \dots, |\lambda_s\rangle, |\lambda_{-1}\rangle, \dots, |\lambda_{-s}\rangle) = (|\lambda_1\rangle, \dots, |\lambda_s\rangle, |\lambda_{-1}\rangle, \dots, |\lambda_{-s}\rangle)\mathbb{S} \quad \#$$

$$= (|\eta_1\rangle, \dots, |\eta_s\rangle, |\eta_{-1}\rangle, \dots, |\eta_{-s}\rangle) \quad \#$$

has real representation matrix $\mathbb{S}$ if $|\lambda\rangle$ and $|\eta\rangle$ belong to the same K equivalence class i.e. $K_\lambda = K_\eta$, if they do not then $\mathbb{S}$ has complex entries.

The complex conjugate of an operator is defined as

$$A^* = KAK, \quad A \in \mathcal{B}(\mathcal{H}^1) \quad \#$$

and the transpose as

$$A^t = (A^*)^\dagger. \quad \#$$

The matrix representation of A^* on a real K basis is

$$KAK(|\lambda_1\rangle, \dots, |\lambda_s\rangle, |\lambda_{-1}\rangle, \dots, |\lambda_{-s}\rangle) = (|\lambda_1\rangle, \dots, |\lambda_s\rangle, |\lambda_{-1}\rangle, \dots, |\lambda_{-s}\rangle)\bar{\mathbb{A}} \quad \#$$

$$A^* = |\chi\rangle\langle\chi|A^*|\chi\rangle\langle\chi| = |\chi\rangle\langle K\chi|AK\chi\rangle\langle\chi| = |\chi\rangle\langle\bar{\mathbb{A}}_\chi|\chi\rangle \quad \#$$

and the representation of A^t on such a basis is

$$A^t(|\lambda_1\rangle, \dots, |\lambda_s\rangle, |\lambda_{-1}\rangle, \dots, |\lambda_{-s}\rangle) = (|\lambda_1\rangle, \dots, |\lambda_s\rangle, |\lambda_{-1}\rangle, \dots, |\lambda_{-s}\rangle)\mathbb{A}^t. \quad \#$$

Transformations of Symplectic Forms

The canonical form can be obtained by transformations of the one particle, $(|\varphi_1\rangle, \dots, |\varphi_s\rangle, |\varphi_{-1}\rangle, \dots, |\varphi_{-s}\rangle)$, of $\mathcal{H}^1$ which have the general effect

$$\begin{aligned} |g\rangle &= \sum_{-s \leq k, l \leq s} g_{kl} |\chi_k \chi_l\rangle = \sum_{-s \leq k, l \leq s} g_{kl} |T\varphi_k T\varphi_l\rangle = \sum_{-s \leq k, l, k', l' \leq s} T_{kk'} T_{ll'} g_{kl} |\varphi_k \varphi_l\rangle \\ &= \sum_{-s \leq k', l' \leq s} |\varphi_{k'} \varphi_{l'}\rangle (T^t g T)_{k'l'}, \end{aligned} \quad \#$$

thus

$$\mathbb{T}_{\chi\varphi}^t \mathbb{G}_\varphi \mathbb{T}_{\chi\varphi} = \mathbb{G}_\chi. \quad \#$$

One also has

$$\mathbb{T}_{\varphi\chi}^t \mathbb{G}_\chi \mathbb{T}_{\varphi\chi} = \mathbb{G}_\varphi \quad \#$$

thus

$$\mathbb{T}_{\chi\varphi}^t \mathbb{T}_{\varphi\chi}^t \mathbb{G}_\chi \mathbb{T}_{\varphi\chi} \mathbb{T}_{\chi\varphi} = \mathbb{G}_\chi. \quad \#$$

Transformations between bases can be chosen to be non singular so

$$\mathbb{T}_{\chi\varphi} = (\mathbb{T}_{\varphi\chi})^{-1}. \quad \#$$

Hence in the standard basis

$$\mathbb{G}_\eta = \begin{pmatrix} \mathbb{O} & -\mathbb{I}_s \\ \mathbb{I}_s & \mathbb{O} \end{pmatrix} \quad \#$$

and in the canonical basis

$$\mathbb{G}_\psi = \begin{pmatrix} \mathbb{O} & -\mathbb{C} \\ \mathbb{C} & \mathbb{O} \end{pmatrix}, \quad \#$$

where

$$\mathbb{G}_{\psi jk} = g(\psi_j, \psi_k); j, k = 1, \dots, s, -1, \dots, -s \quad \#$$

and thus

$$\mathbb{C} = \begin{pmatrix} c_1 & 0 & \dots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & c_s \end{pmatrix}. \quad \#$$

The transformation between the canonical and the standard basis is

$$\begin{aligned} \begin{pmatrix} \mathbb{C}^{-\frac{1}{2}} & \mathbb{O} \\ \mathbb{O} & \mathbb{C}^{-\frac{1}{2}} \end{pmatrix} \begin{pmatrix} \mathbb{O} & -\mathbb{C} \\ \mathbb{C} & \mathbb{O} \end{pmatrix} \begin{pmatrix} \mathbb{C}^{-\frac{1}{2}} & \mathbb{O} \\ \mathbb{O} & \mathbb{C}^{-\frac{1}{2}} \end{pmatrix} = \\ \begin{pmatrix} \mathbb{O} & -\mathbb{C}^{\frac{1}{2}} \\ \mathbb{C}^{\frac{1}{2}} & \mathbb{O} \end{pmatrix} \begin{pmatrix} \mathbb{C}^{-\frac{1}{2}} & \mathbb{O} \\ \mathbb{O} & \mathbb{C}^{-\frac{1}{2}} \end{pmatrix} = \begin{pmatrix} \mathbb{O} & -\mathbb{I}_s \\ \mathbb{I}_s & \mathbb{O} \end{pmatrix} \end{aligned} \quad \#$$

thus

$$\mathbb{T}_{\eta\psi} = \begin{pmatrix} \mathbb{C}^{-\frac{1}{2}} & \mathbb{O} \\ \mathbb{O} & \mathbb{C}^{-\frac{1}{2}} \end{pmatrix}. \quad \#$$

The wavefunction associated with $|g\rangle$ is given by

$$g(\mathbf{x}_1, \mathbf{x}_2) = \langle \mathbf{x}_1 \mathbf{x}_2 | g \rangle = \sum_{1 \leq j \leq s} \langle \mathbf{x}_1 \mathbf{x}_2 | \eta_j \eta_{-j} \rangle \quad \#$$

$$= \sum_{1 \leq j \leq s} |\eta_j(\mathbf{x}_1) \eta_{-j}(\mathbf{x}_2)\rangle = \sum_{1 \leq j \leq s} c_j |\psi_j(\mathbf{x}_1) \psi_{-j}(\mathbf{x}_2)\rangle, \quad \#$$

thus

$$c_j = \frac{1}{\sqrt{2}} \int \overline{\{\psi_j(\mathbf{x}_1) \psi_{-j}(\mathbf{x}_2) - \psi_j(\mathbf{x}_2) \psi_{-j}(\mathbf{x}_1)\}} g(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2 \quad \#$$

and

$$\int \varphi_a(\mathbf{x}_1) g(\mathbf{x}_1, \mathbf{x}_2) \varphi_b(\mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2 = g(\varphi_a, \varphi_b), \quad \#$$

while basis transformations give

$$g(\mathbf{x}_1 \mathbf{x}_2) = \sum_{-s \leq j \leq s} c_j |T\varphi_j(\mathbf{x}_1) T\varphi_{-j}(\mathbf{x}_2)\rangle = \sum_{-s \leq j, k, l \leq s} T_{jk} T_{-jl} c_j |\varphi_k(\mathbf{x}_1) \varphi_l(\mathbf{x}_2)\rangle.$$

The amplitudes, $\langle \mathbf{x}_1 | \varphi_j \rangle$, of the basis elements $|\varphi_j\rangle$ are the "matrix elements" of the transformation $\Upsilon : (|\mathbf{x}_1\rangle) \rightarrow (|\varphi_1\rangle \cdots |\varphi_s\rangle |\varphi_{-1}\rangle \cdots |\varphi_{-s}\rangle)$ from a discrete basis to a continuous generalized one i.e.

$$|\mathbf{x}_1\rangle = \sum_{-s \leq j \leq s} |\varphi_j\rangle \langle \varphi_j | \mathbf{x}_1 \rangle \equiv \sum_{-s \leq j \leq s} |\varphi_j\rangle \overline{\Upsilon_j(\mathbf{x}_1)} \quad \#$$

giving

$$(\Upsilon^\dagger g \tilde{\Upsilon})(\mathbf{x}_1, \mathbf{x}_2) = g(\mathbf{x}_1, \mathbf{x}_2) = \sum_{-s \leq j, k \leq s} \overline{\Upsilon_j(\mathbf{x}_1)} g_{jk} \overline{\Upsilon_k(\mathbf{x}_2)}. \quad \#$$

The inverse transformation is given by

$$|\varphi_j\rangle = \int |\mathbf{x}_1\rangle \langle \mathbf{x}_1 | \varphi_j \rangle d\mathbf{x}_1 \equiv \int |\mathbf{x}_1\rangle \Upsilon_j(\mathbf{x}_1) d\mathbf{x}_1 \quad \#$$

leading to

$$(\Upsilon^t g \Upsilon)_{jk} = \int \Upsilon_j(\mathbf{x}_1) g(\mathbf{x}_1 \mathbf{x}_2) \Upsilon_k(\mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2. \quad \#$$

Symplectic Group

The symplectic group $Sp(\mathcal{H}^1) \equiv Sp(2s, \mathbb{C})$ is defined as the set of invertible transformations S that satisfy

$$g(S\varphi_1, S\varphi_2) = g(\varphi_1, \varphi_2) \forall \varphi_1, \varphi_2 \in \mathcal{H}^1 \quad \#$$

if

$$\text{rank}\{g\} = s. \quad \#$$

If

$$\text{rank}\{g\} = \rho < s \quad \#$$

then this defines the subgroup $Sp(2\rho, \mathbb{C})$ of $Sp(2s, \mathbb{C})$. As

$$g(\varphi_1, \varphi_2) = \mathbf{v}_{1\lambda}^t \mathbb{G}_\lambda \mathbf{v}_{2\lambda}, \quad \#$$

where

$$|v\rangle = (|\lambda_1\rangle, \dots, |\lambda_s\rangle, |\lambda_{-1}\rangle, \dots, |\lambda_{-s}\rangle) \mathbf{v} \quad \#$$

one has

$$g(S\varphi_1, S\varphi_2) = \mathbf{v}_{1\lambda}^t \mathbb{S}_\lambda^t \mathbb{G}_\lambda \mathbb{S}_\lambda \mathbf{v}_{2\lambda} = \mathbf{v}_{1\lambda}^t \mathbb{G}_\lambda \mathbf{v}_{2\lambda} \quad \forall \mathbf{v}_{1\lambda}, \mathbf{v}_{2\lambda} \in l_2, \quad \#$$

which shows that

$$\mathbb{S}_\lambda^t \mathbb{G}_\lambda \mathbb{S}_\lambda = \mathbb{G}_\lambda \quad \#$$

and

$$\mathbb{S}_\lambda^t \mathbb{T}_{\lambda\eta}^t \mathbb{G}_\eta \mathbb{T}_{\lambda\eta} \mathbb{S}_\lambda = \mathbb{G}_\lambda = \mathbb{T}_{\lambda\eta}^t \mathbb{G}_\eta \mathbb{T}_{\lambda\eta}, \quad \#$$

giving

$$\begin{aligned} \mathbb{T}_{\lambda\eta}^{-t} \mathbb{S}_\lambda^t \mathbb{T}_{\eta\lambda}^t \mathbb{G}_\eta \mathbb{T}_{\eta\lambda} \mathbb{S}_\lambda \mathbb{T}_{\lambda\eta}^{-1} &= \mathbb{G}_\eta \\ &\equiv [\mathbb{T}_{\eta\lambda} \mathbb{S}_\lambda \mathbb{T}_{\lambda\eta}^{-1}]^t \mathbb{G}_\eta^t [\mathbb{T}_{\eta\lambda} \mathbb{S}_\lambda \mathbb{T}_{\lambda\eta}^{-1}] = \mathbb{G}_\eta, \end{aligned} \quad \#$$

which shows that $\mathbb{S}_\lambda$ and $\mathbb{S}_\eta$ are conjugate representations of $Sp(2s, \mathbb{C})$.

Invariance Group

One can observe from the preceding that the invariance group of $|g\rangle$ is $Sp(2\rho, \mathbb{C})$, which is a subgroup of $GL(\mathcal{H}^1)$ that has an irreducible representation on $\mathcal{H}^2$. The hilbert space $\mathcal{H}^2$ is not a homogeneous space for $GL(\mathcal{H}^1)$, but each orbit does generate $\mathcal{H}^2$ and each vector in $\mathcal{H}^2$ is cyclic for this group and thus all elements of $\mathcal{H}^2$ i.e. geminals are generated by $GL(\mathcal{H}^1)$ acting on a reference geminal by the taking closed linear span of the orbit produced by the action of the coset

$$\frac{GL(\mathcal{H}^1)}{Sp(2\rho, \mathbb{C})} \equiv \frac{GL(2s, \mathbb{C})}{Sp(2\rho, \mathbb{C})} \quad \#$$

on the reference geminal

$$|g_\rho\rangle = \sum_{1 \leq j \leq s} c_{\rho j} |\psi_{\rho j} \psi_{\rho-j}\rangle. \quad \#$$

In detail this orbit is given by

$$\begin{aligned} \frac{GL(2s, \mathbb{C})}{Sp(2\rho, \mathbb{C})} Sp(2\rho, \mathbb{C}) |g_\rho\rangle &= \{|g_\rho(\gamma)\rangle = \gamma |g_\rho\rangle | \gamma \in GL(2s, \mathbb{C})\} \\ &= \left\{ |g_\rho(c)\rangle = ch |g_\rho\rangle \middle| c \in \frac{GL(2s, \mathbb{C})}{Sp(2\rho, \mathbb{C})}, h \in Sp(2\rho, \mathbb{C}) \right\}. \end{aligned} \quad \#$$

The set $\{|g_\rho(c)\rangle\}$ is a representation of the homogeneous coset space $\frac{GL(2s, \mathbb{C})}{Sp(2\rho, \mathbb{C})}$ as a non linear manifold contained in $\mathcal{H}^2$.

Lie Algebra

The Lie algebra of $Sp(2\rho, \mathbb{C})$ is generated by considering one parameter subgroups $\{S(\tau) | \tau \in (-\infty, \infty)\}$ of $Sp(2\rho, \mathbb{C})$

$$S^t(\tau) g S(\tau) = g \equiv g S(\tau) = S(\tau)^{-t} g \quad \#$$

then

$$S(\tau) = e^{s\tau} = I + s\tau + \dots \quad \#$$

and to first order

$$gs = -s^t g = (gs)^t. \quad \#$$

This gives the structure of the Lie algebra matrices in the standard representation as

$$\begin{pmatrix} \mathbb{O} & -\mathbb{I} \\ \mathbb{I} & \mathbb{O} \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} -c & -d \\ a & b \end{pmatrix} = \begin{pmatrix} -c^t & a^t \\ -d^t & b^t \end{pmatrix} \quad \#$$

thus

$$s_\eta = \begin{pmatrix} a & b \\ c & -a^t \end{pmatrix}; b = b^t, c = c^t. \quad \#$$

One can observe that

$$\begin{pmatrix} \mathbb{O} & -\mathbb{I} \\ \mathbb{I} & \mathbb{O} \end{pmatrix} \begin{pmatrix} a & b \\ c & -a^t \end{pmatrix} = \begin{pmatrix} -c & a^t \\ a & b \end{pmatrix}, \quad \#$$

which is a complex symmetric matrix. The inverse of this transformation is

$$\begin{pmatrix} \mathbb{O} & -\mathbb{I} \\ \mathbb{I} & \mathbb{O} \end{pmatrix} \begin{pmatrix} \mathbb{O} & \mathbb{I} \\ -\mathbb{I} & \mathbb{O} \end{pmatrix} = \begin{pmatrix} \mathbb{I} & \mathbb{O} \\ \mathbb{O} & \mathbb{I} \end{pmatrix} \quad \#$$

and

$$\begin{pmatrix} -c & a^t \\ a & b \end{pmatrix} \begin{pmatrix} \mathbb{O} & \mathbb{I} \\ -\mathbb{I} & \mathbb{O} \end{pmatrix} = \begin{pmatrix} -a^t & -c \\ -b & a \end{pmatrix} \quad \#$$

The canonical representation also has the same structure i.e.

$$\begin{pmatrix} \mathbb{C}^{\frac{1}{2}} & \mathbb{O} \\ \mathbb{O} & \mathbb{C}^{\frac{1}{2}} \end{pmatrix} \begin{pmatrix} a & b \\ c & -a^t \end{pmatrix} \begin{pmatrix} \mathbb{C}^{-\frac{1}{2}} & \mathbb{O} \\ \mathbb{O} & \mathbb{C}^{-\frac{1}{2}} \end{pmatrix} = \begin{pmatrix} a & b \\ c & -a^t \end{pmatrix}. \quad \#$$

Fock Space and Fock Algebra

The second quantisation map $\Phi : \mathcal{H}^1 \rightarrow \mathcal{B}(\mathcal{H}_F)$ is defined as

$$\Phi \left(\sum_{1 \leq j \leq r} c_j \varphi_j \right) = \sum_{1 \leq j \leq r} \bar{c}_j \Phi(\varphi_j) \quad \#$$

and the creators and annihilators associated with the basis $\{\varphi_j\}$ as

$$\begin{aligned} a_j^\dagger &= \Phi(\varphi_j)^\dagger \\ a_j &= \Phi(\varphi_j). \end{aligned} \quad \#$$

The space-spin field operators are defined in terms of distributions as

$$\Phi(\mathbf{x}) = \Phi(\delta_{\mathbf{x}}) = \sum_{1 \leq j \leq r} \overline{\varphi_j(\mathbf{x})} a_j$$

$$\Phi(\mathbf{x})^\dagger = \Phi(\delta_{\mathbf{x}})^\dagger = \sum_{1 \leq j \leq r} \varphi_j(\mathbf{x}) a_j^\dagger \quad \#$$

leading to the geminal expressions

$$|g\rangle = \sum_{1 \leq j \leq s} c_j a_j^\dagger a_{-j}^\dagger |\phi\rangle = \sum_{1 \leq j \leq s} \Phi(\eta_j)^\dagger \Phi(\eta_{-j})^\dagger |\phi\rangle. \quad \#$$

and

$$|g\rangle = \int g(\mathbf{x}_1, \mathbf{x}_2) \Phi(\mathbf{x}_1)^\dagger \Phi(\mathbf{x}_2)^\dagger d\mathbf{x}_1 d\mathbf{x}_2 |\phi\rangle \quad \#$$

Representations

The Lie algebra generated by

$$q_{jk} = (c_j a_j^\dagger a_k - \sigma(j)\sigma(k) c_k a_{-k}^\dagger a_{-j}); 1 \leq |j| \leq |k| \leq s, j \neq -k \quad \#$$

$$q_{j-j} = a_j^\dagger a_{-j}; -s \leq j \leq s \quad \#$$

form a reducible representation $R(sp(2s, \mathbb{C}))$ of the Lie algebra $sp(2s, \mathbb{C})$ on the Fock space, $\mathcal{H}_F$, that can be decomposed into the direct sum of irreps

$$R(sp(2s, \mathbb{C})) = \bigoplus_{0 \leq N \leq 2s} R_N(sp(2s, \mathbb{C})), \quad \#$$

where $\{R_N(sp(2s, \mathbb{C})) | 0 \leq N \leq r\}$ are the irreducible representations carried by $\mathcal{H}^N$.

In general

$$\Phi(T\varphi_j)^\dagger = \Phi\left(\sum_{1 \leq j' \leq r} T_{jj'} \varphi_{j'}\right)^\dagger = \sum_{1 \leq j' \leq r} T_{jj'} \Phi(\varphi_{j'})^\dagger = \sum_{1 \leq j' \leq r} T_{jj'} a_{j'}^\dagger \quad \#$$

leading to

$$\Phi(T\varphi_j)^\dagger \Phi(T\varphi_k) = \sum_{1 \leq j', k' \leq r} T_{jj'} \bar{T}_{kk'} a_{j'}^\dagger a_{k'} = \sum_{1 \leq j', k' \leq r} (T \otimes \bar{T})_{jk'j'k'} a_{j'}^\dagger a_{k'} \quad \#$$

and

$$T a_j^\dagger a_k T^\dagger = \Phi(T\varphi_j)^\dagger \Phi(T\varphi_k) = \sum_{1 \leq j', k' \leq r} T_{jj'} a_{j'}^\dagger a_{k'} T_{k'k}^\dagger. \quad \#$$

Lie Algebra Bases

Gauss Decomposition Basis

If the geminal is non degenerate i.e. $c_j \neq c_k$ the Lie algebra, $gl(2s, \mathbb{C})$, of $GL(2s, \mathbb{C})$ is spanned by the basis elements

$$q_{jk}^\dagger = (c_j a_k^\dagger a_j - \sigma(j)\sigma(k) c_k a_{-j}^\dagger a_{-k}); -s \leq j < k \leq s, |j| \neq |k|$$

$$n_j = c_j (a_j^\dagger a_j + a_{-j}^\dagger a_{-j}); 1 \leq j \leq s$$

$$q_{jk} = (c_j a_j^\dagger a_k - \sigma(j)\sigma(k) c_k a_{-k}^\dagger a_{-j}); -s \leq j < k \leq s; j \neq -k$$

$$q_{j-j} = a_j^\dagger a_{-j}; -s \leq j \leq s, \quad \#$$

which satisfy the indexing symmetry

$$q_{jk} = -\sigma(j)\sigma(k)q_{-k-j}; 1 \leq j, k \leq s.$$

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In particular

$$q_{jj} = c_j(a_j^\dagger a_j - a_{-j}^\dagger a_{-j}); 1 \leq j \leq s$$

and one has the adjoint relationships

$$q_{j-j} = q_{-jj}^\dagger, -s \leq j \leq s$$

$$q_{jj} = q_{jj}^\dagger, n_j^\dagger = n_j, 1 \leq j \leq s.$$

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The transformation between the "off diagonal" q operators and the "a" operators can be succintly displayed for $-s \leq j < k \leq s, |k| \neq |j|$ as

$$\begin{pmatrix} q_{jk} \\ q_{j-k} \\ q_{-jk} \\ q_{-j-k} \\ q_{jk}^\dagger \\ q_{j-k}^\dagger \\ q_{-jk}^\dagger \\ q_{-j-k}^\dagger \end{pmatrix} = \begin{pmatrix} c_j & 0 & \cdots & \cdots & \cdots & \cdots & 0 & -c_k \\ 0 & c_j & 0 & \cdots & \cdots & 0 & c_k & 0 \\ \vdots & 0 & c_j & 0 & 0 & c_k & 0 & \vdots \\ \vdots & \vdots & 0 & c_j & -c_k & 0 & \vdots & \vdots \\ \vdots & \vdots & 0 & -c_k & c_j & 0 & \vdots & \vdots \\ \vdots & 0 & c_k & 0 & 0 & c_j & 0 & \vdots \\ 0 & c_k & 0 & \cdots & \cdots & 0 & c_j & 0 \\ -c_k & 0 & \cdots & \cdots & \cdots & \cdots & 0 & c_j \end{pmatrix} \begin{pmatrix} a_j^\dagger a_k \\ a_j^\dagger a_{-k} \\ a_{-j}^\dagger a_k \\ a_{-j}^\dagger a_{-k} \\ a_k^\dagger a_j \\ a_{-k}^\dagger a_j \\ a_k^\dagger a_{-j} \\ a_{-k}^\dagger a_{-j} \end{pmatrix}.$$

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Commutation Relationships

This basis produces a Gauss Decomposition

$$L_{-1} \oplus L_0 \oplus L_{+1}$$

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where

$$L_{+1} = \text{linspan}\{q_{jk}^\dagger = (c_j a_k^\dagger a_j - \sigma(j)\sigma(k)c_k a_{-j}^\dagger a_{-k}); -s \leq j < k \leq s, |j| \neq |k|\}$$

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$$L_0 = \text{linspan}\{q_{j-j} = a_j^\dagger a_{-j}; -s \leq j \leq s; a_j^\dagger a_j - a_{-j}^\dagger a_{-j}, 1 \leq j \leq s\}$$

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$$L_{-1} = \text{linspan}\{q_{jk} = (c_j a_j^\dagger a_k - \sigma(j)\sigma(k)c_k a_{-k}^\dagger a_{-j}); -s \leq j < k \leq s; |j| \neq |k|\}$$

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as

$$[\mathbf{q}, \mathbf{q}] \subset L_{-1} \oplus L_0$$

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$$[\mathbf{q}^\dagger, \mathbf{q}^\dagger] \subset L_{+1} \oplus L_0$$

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$$[\mathbf{q}, \mathbf{q}_0] \subset L_{-1}$$

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$$[\mathbf{q}^\dagger, \mathbf{q}_0] \subset L_{+1}$$

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$$[\mathbf{q}_0, \mathbf{q}_0] \subset L_0$$

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Action on Geminal

The action of these operators on a canonical GSO is

$$\sum_{1 \leq k < l \leq s} \{z_{kl} q_{kl}^\dagger + z_{-kl} q_{-kl}^\dagger + z_{k-l} q_{k-l}^\dagger + z_{-k-l} q_{-k-l}^\dagger\} |\psi_j\rangle$$

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$$z_{kl} = -\sigma(k)\sigma(l)z_{-l-k}; -s \leq k, l \leq s, z_{kl} = 0, |k| = |l| \quad \#$$

The general linear group $GL(\mathcal{H}^1)$ and can thus be parameterized by

$$\gamma(\boldsymbol{\alpha}, \mathbf{z}, \boldsymbol{\beta}, \mathbf{w}) = c(\boldsymbol{\alpha}, \mathbf{z})h(\boldsymbol{\beta}, \mathbf{w}) = c_1(\boldsymbol{\alpha}_{\text{re}}, \boldsymbol{\alpha}_{\text{im}})\tau_+(\mathbf{z})\tau_-(\mathbf{w})h_1(\boldsymbol{\beta}) \quad \#$$

where

$$\begin{aligned} c(\boldsymbol{\alpha}, \mathbf{z}) &= c_1(\boldsymbol{\alpha}_{\text{re}}, \boldsymbol{\alpha}_{\text{im}})\tau_+(\mathbf{z}) \\ h(\boldsymbol{\beta}, \mathbf{w}) &= \tau_-(\mathbf{w})h_1(\boldsymbol{\beta}) \end{aligned} \quad \#$$

and the parameterization is produced by the coset representatives

$$\begin{aligned} \tau_+(\mathbf{z}) &= \exp \left\{ \sum_{\substack{-s \leq j < k \leq s \\ |j| \neq |k|}} z_{jk} q_{jk}^\dagger \right\}; -s \leq j < k \leq s, z_{jk} \in \mathbb{C} \\ c_{1i}(\boldsymbol{\alpha}_{\text{im}}) &= \exp \left\{ i \sum_{1 \leq j \leq s} \alpha_{\text{im}j} \mathbf{n}_j \right\}; 1 \leq j \leq s, \alpha_{\text{im}j} \in \mathbb{R} \\ c_{1r}(\boldsymbol{\alpha}_{\text{re}}) &= \exp \left\{ \sum_{1 \leq j \leq s} \alpha_{\text{re}j} \mathbf{n}_j \right\}; 1 \leq j \leq s, \alpha_{\text{re}j} \in \mathbb{R}, \end{aligned} \quad \#$$

where

$$c_1(\boldsymbol{\alpha}_{\text{re}}, \boldsymbol{\alpha}_{\text{im}}) = c_{1i}(\boldsymbol{\alpha}_{\text{im}})c_{1r}(\boldsymbol{\alpha}_{\text{re}}) \quad \#$$

and the invariant subgroup

$$\begin{aligned} \tau_-(\mathbf{w}) &= \exp \left\{ \sum_{\substack{-s \leq j < k \leq s \\ |j| \neq |k|}} w_{jk} q_{jk} \right\}; -s \leq j < k \leq s, w_{jk} \in \mathbb{C} \\ h_1(\boldsymbol{\beta}) &= \exp \left\{ \sum_{1 \leq j \leq s} (\beta_{jj}(\hat{n}_j - \hat{n}_{-j}) + \beta_{j-j} a_j^\dagger a_{-j} + \beta_{-jj} a_{-j}^\dagger a_j) \right\}, \beta_{jk} \in \mathbb{C}. \end{aligned} \quad \#$$

The basis $\{\mathbf{n}_j | 1 \leq j \leq s\}$ for the coset representatives $c_1(\boldsymbol{\alpha}_{\text{re}}, \boldsymbol{\alpha}_{\text{im}})$ can be transformed to

$$\{\mathcal{N}_1, \mathbf{m}_j = \mathbf{n}_j - \mathbf{n}_{j+1} | 1 \leq j \leq s-1\}, \quad \#$$

which allows one to perform the factorization

$$GL(2s, \mathbb{C}) = GL(1, \mathbb{C}) \times SL(2s, \mathbb{C}) \quad \#$$

and set $c(\boldsymbol{\alpha}, \mathbf{z})$ to

$$c(\tilde{\boldsymbol{\alpha}}, \mathbf{z}) = s(\tilde{\alpha}_{1r})p(\tilde{\alpha}_{1i})d(\tilde{\boldsymbol{\alpha}})\tau_+(\mathbf{z}), \quad \#$$

where

$$s(\tilde{\alpha}_{1r})p(\tilde{\alpha}_{1i})d(\tilde{\alpha}) = e^{\tilde{\alpha}_{1r}\mathcal{N}_1}e^{i\tilde{\alpha}_{1i}\mathcal{N}_1}\exp\left\{\sum_{1\leq j\leq s-1}\tilde{\alpha}_j\mathfrak{m}_j\right\} \quad \#$$

$$\equiv s(\tilde{\alpha}_{1r})p(\tilde{\alpha}_{1i})d(\tilde{\alpha}); 2\leq j\leq s, \tilde{\alpha}_j\in\mathbb{C}, \tilde{\alpha}_{1r}, \tilde{\alpha}_{1i}\in\mathbb{R} \quad \#$$

and $e^{\tilde{\alpha}_{1r}\mathcal{N}_1}$ and $e^{i\tilde{\alpha}_{1i}\mathcal{N}_1}$ are overall scaling and phase factors.

$$\exp\left\{\sum_{1\leq j\leq s-1}\tilde{\alpha}_j\mathfrak{m}_j\right\}\exp\left\{\sum_{\substack{-s\leq j<k\leq s \\ |j|\neq|k|}}z_{jk}q_{jk}^\dagger\right\}|g_\rho\rangle = \sum_{1\leq j\leq s}|\Upsilon\eta_{\rho j}\Upsilon\eta_{-\rho j}\rangle, \Upsilon\in\frac{SL(2s,\mathbb{C})}{Sp(2\rho,\mathbb{C})} \quad \#$$

The operators $\{q_{jk}^\dagger \mid -s\leq j<k\leq s, |j|\neq|k|\}$ and $\{q_{jk} \mid -s\leq j<k\leq s, |j|\neq|k|\}$ satisfy the same commutation relationships and are representatives of the coset $\frac{Sp(2s,\mathbb{C})}{SL(2)^s}$ where

$$Sp(2s,\mathbb{C}) = \frac{Sp(2s,\mathbb{C})}{SL(2)^s}SL(2)^s. \quad \#$$

(It is important to note that the in representation produced by the non degenerate q operators there is *no* representation of the subgroup $Sp(2s,\mathbb{C})$). One can observe from the preceeding that

$$SL(2s,\mathbb{C}) = \frac{Sp(2s,\mathbb{C})}{SL(2)^s} \frac{Sp(2s,\mathbb{C})}{SL(2)^s} SL(2)^s \cong \frac{Sp(2s,\mathbb{C})}{SL(2)^s} SL(2)^s \frac{Sp(2s,\mathbb{C})}{SL(2)^s}. \quad \#$$

One thus also has that

$$SL(2s,\mathbb{C}) = Sp(2s,\mathbb{C})SL(2)^sSp(2s,\mathbb{C}), \quad \#$$

which is the complexification of

$$SU^*(2s)SU(2)^sSU^*(2s). \quad \#$$

$$\lceil \mathfrak{B}_1(\mathbb{H}^1) \rceil \cong SU(2) \equiv SU(2,\mathbb{C}) \quad \#$$

$$\mathbb{H}^1 = \left\{ \left(\begin{array}{cc} x & y \\ -\bar{y} & \bar{x} \end{array} \right) \middle| x, y \in \mathbb{C} \right\} \quad \#$$

$$su^*(2) = \mathbb{H}^1 \rfloor \quad \#$$

Hence

$$\exp\left\{\sum_{\substack{-s\leq j<k\leq s \\ |j|\neq|k|}}z_{jk}q_{jk}^\dagger\right\}(|\psi_1\rangle\cdots|\psi_s\rangle, |\psi_{-1}\rangle\cdots|\psi_{-s}\rangle) \quad \#$$

$$= (|\psi_1\rangle\cdots|\psi_s\rangle, |\psi_{-1}\rangle\cdots|\psi_{-s}\rangle)\mathbb{W}(\mathbb{Z}), \quad \#$$

where $\mathbb{W}(\mathbb{Z})$ is a matrix representative of $\frac{Sp(2s,\mathbb{C})}{SL(2)^s}$ wrt the basis $\{|\psi_j\rangle \mid -s\leq j\leq s\}$ that depends on $\mathbb{Z}$.

Cartan Decomposition

In order to obtain the Cartan decomposition of the Lie algebra one considers the basis

$$u_{jk} = c_k a_j^\dagger a_k + \sigma(j)\sigma(k)c_j a_{-k}^\dagger a_{-j}; -s \leq j < k \leq s, |j| \neq |k| \quad \#$$

$$m_j = c_j(a_j^\dagger a_j + a_{-j}^\dagger a_{-j}) - c_{j+1}(a_{j+1}^\dagger a_{j+1} + a_{-(j+1)}^\dagger a_{-(j+1)}); 1 \leq j \leq s-1 \quad \#$$

$$v_{jk} = c_k a_{-k}^\dagger a_{-j} - \sigma(j)\sigma(k)c_j a_j^\dagger a_k; \quad \#$$

$$v_{j-j} = a_j^\dagger a_{-j}; -s \leq j \leq s \quad \#$$

$$v_{jj} = c_j(a_j^\dagger a_j - a_{-j}^\dagger a_{-j}); 1 \leq j \leq s \quad \#$$

The transformation between the "off diagonal" (q, u) operators and the "a" operators can be succinctly displayed for $-s \leq j < k \leq s, |k| \neq |j|$ as

$$\begin{pmatrix} q_{jk} \\ q_{j-k} \\ q_{-jk} \\ q_{-j-k} \\ u_{jk} \\ u_{j-k} \\ u_{-jk} \\ u_{-j-k} \end{pmatrix} = \begin{pmatrix} c_j & 0 & \cdots & \cdots & \cdots & \cdots & 0 & -c_k \\ 0 & c_j & 0 & \cdots & \cdots & 0 & c_k & 0 \\ \vdots & 0 & c_j & 0 & 0 & c_k & 0 & \vdots \\ \vdots & \vdots & 0 & c_j & -c_k & 0 & \vdots & \vdots \\ \vdots & \vdots & 0 & c_k & c_j & 0 & \vdots & \vdots \\ \vdots & 0 & -c_k & 0 & 0 & c_j & 0 & \vdots \\ 0 & -c_k & 0 & \cdots & \cdots & 0 & c_j & 0 \\ c_k & 0 & \cdots & \cdots & \cdots & \cdots & 0 & c_j \end{pmatrix} \begin{pmatrix} a_j^\dagger a_k \\ a_j^\dagger a_{-k} \\ a_{-j}^\dagger a_k \\ a_{-j}^\dagger a_{-k} \\ a_k^\dagger a_j \\ a_{-k}^\dagger a_j \\ a_k^\dagger a_{-j} \\ a_{-k}^\dagger a_{-j} \end{pmatrix}. \quad \#$$

Action on Geminal

$$u_{kl}|g\rangle = (c_k a_l^\dagger a_k + c_l a_{-k}^\dagger a_{-l}) \sum_{1 \leq j \leq s} c_j |\psi_j \psi_{-j}\rangle = (n_k + n_l) |\psi_l \psi_{-k}\rangle \quad \#$$

$$u_{k-l}|g\rangle = (c_k a_{-l}^\dagger a_k - c_l a_{-k}^\dagger a_l) \sum_{1 \leq j \leq s} c_j |\psi_j \psi_{-j}\rangle = (n_k + n_l) |\psi_{-l} \psi_{-k}\rangle \quad \#$$

$$u_{-k-l}|g\rangle = (c_k a_l^\dagger a_{-k} - c_l a_k^\dagger a_{-l}) \sum_{1 \leq j \leq s} c_j |\psi_j \psi_{-j}\rangle = (n_k + n_l) |\psi_k \psi_l\rangle \quad \#$$

$$u_{-k-l}|g\rangle = (c_k a_{-l}^\dagger a_{-k} + c_l a_k^\dagger a_l) \sum_{1 \leq j \leq s} c_j |\psi_j \psi_{-j}\rangle = (n_k + n_l) |\psi_k \psi_{-l}\rangle \quad \#$$

Orbits of $GL(2s, \mathbb{C})$ generated by $|g_\rho\rangle$

The orbits of $GL(2s, \mathbb{C})$ generated by $|g_\rho\rangle$ are given by

$$\begin{aligned} \gamma(\alpha, \mathbf{z}, \beta, \mathbf{w})|g_\rho\rangle &= c(\alpha, \mathbf{z})h(\beta, \mathbf{w})|g_\rho\rangle = c_1(\tilde{\alpha})\tau_+(\mathbf{z})\tau_-(\mathbf{w})h_1(\beta)|g_\rho\rangle \\ &= c_1(\tilde{\alpha})\tau_+(\mathbf{z})|g_\rho\rangle \end{aligned} \quad \#$$

which gives a representation of the coset space $\frac{GL(1, \mathbb{C}) \times SL(2s, \mathbb{C})}{sp(2s, \mathbb{C})}$ as the nonlinear manifold

$$|g_\rho(\tilde{\alpha}, \mathbf{z})\rangle = c_1(\tilde{\alpha})\tau_+(\mathbf{z})|g_\rho\rangle = s(\tilde{\alpha}_{1r})p(\tilde{\alpha}_{1i})d(\tilde{\alpha})\tau_+(\mathbf{z})|g_\rho\rangle. \quad \#$$

One can restrict the orbit to lie in unit ball of $\mathcal{H}^2$ if $\| |g_\rho\rangle \| = 1$ by setting

$$\tilde{\alpha}_{1r}(\tilde{\alpha}, \mathbf{z}) = -\frac{1}{4} \ln \{ \langle g_\rho(\tilde{\alpha}, \mathbf{z}) | g_\rho(\tilde{\alpha}, \mathbf{z}) \rangle \} = -\frac{1}{4} \ln \{ \langle g_\rho | d(\tilde{\alpha})\tau_+(\mathbf{z})^\dagger \tau_+(\mathbf{z})d(\tilde{\alpha})g_\rho \rangle \} \quad \#$$

Antisymmetrized Geminal Powers (AGP's)

A reference $2N$ -electron AGP state in $\mathcal{H}^{2N}$ is defined by

$$|g_\rho^N\rangle = \sum_{1 \leq j_1 < j_2 < \dots < j_N \leq S} c_{\rho j_1} \dots c_{\rho j_N} |\psi_{j_1} \psi_{-j_1} \dots \psi_{j_N} \psi_{-j_N}\rangle \quad \#$$

and the wavefunction associated with this state is

$$g_\rho^N(\mathbf{x}_1 \dots \mathbf{x}_{2N}) = \langle \mathbf{x}_1 \dots \mathbf{x}_{2N} | g_\rho^N \rangle = Pf(g_\rho)(\mathbf{x}_1 \dots \mathbf{x}_{2N}). \quad \#$$

The CI expansion is given by

$$Pf(Y^\dagger g \bar{Y}) = \sum \det(\bar{Y})_p Pf(g_\rho)_p \quad \#$$

This reference state can be expressed in operator form as

$$\begin{aligned} |g_\rho^N\rangle &= \sum_{1 \leq j_1 < j_2 < \dots < j_N \leq S} c_{\rho j_1} \dots c_{\rho j_N} a_{j_1}^\dagger a_{-j_1}^\dagger \dots a_{j_N}^\dagger a_{-j_N}^\dagger |\phi\rangle \\ &= \frac{1}{2^N N!} \int g_\rho(\mathbf{x}_1, \mathbf{x}_2) \dots g_\rho(\mathbf{x}_{2N-1}, \mathbf{x}_{2N}) \Phi(\mathbf{x}_1)^\dagger \dots \Phi(\mathbf{x}_{2N})^\dagger d\mathbf{x}_1 \dots d\mathbf{x}_{2N} |\phi\rangle \\ &= \frac{1}{2^N N!} \int Pf(g_\rho)(\mathbf{x}_1 \dots \mathbf{x}_{2N}) \Phi(\mathbf{x}_1)^\dagger \dots \Phi(\mathbf{x}_{2N})^\dagger d\mathbf{x}_1 \dots d\mathbf{x}_{2N} |\phi\rangle. \end{aligned} \quad \#$$

The norm squared of this state is

$$\begin{aligned} \langle g_\rho^N | g_\rho^N \rangle &= \left(\frac{1}{2^N N!} \right)^2 \int \overline{Pf(g_\rho)(\mathbf{x}_1 \dots \mathbf{x}_{2N})} Pf(g_\rho)(\mathbf{x}_1 \dots \mathbf{x}_{2N}) d\mathbf{x}_1 \dots d\mathbf{x}_{2N} \\ &= \left(\frac{1}{2^N N!} \right)^2 \int |Pf(g_\rho)(\mathbf{x}_1 \dots \mathbf{x}_{2N})|^2 d\mathbf{x}_1 \dots d\mathbf{x}_{2N} \\ &= \left(\frac{1}{2^N N!} \right)^2 \int |\det(g_\rho)(\mathbf{x}_1 \dots \mathbf{x}_{2N})| d\mathbf{x}_1 \dots d\mathbf{x}_{2N}, \end{aligned} \quad \#$$

where

$$\det(g_\rho)(\mathbf{x}_1 \dots \mathbf{x}_{2N}) = \det \begin{pmatrix} g_\rho(\mathbf{x}_1, \mathbf{x}_1) & \dots & g_\rho(\mathbf{x}_1, \mathbf{x}_{2N}) \\ \vdots & \ddots & \vdots \\ g_\rho(\mathbf{x}_{2N}, \mathbf{x}_1) & \dots & g_\rho(\mathbf{x}_{2N}, \mathbf{x}_{2N}) \end{pmatrix}. \quad \#$$

This should be compared to

$$\Phi(\mathbf{x}_1 \dots \mathbf{x}_{2N}) = \det \begin{pmatrix} \varphi_1(\mathbf{x}_1) & \dots & \varphi_{2N}(\mathbf{x}_1) \\ \vdots & \ddots & \vdots \\ \varphi_1(\mathbf{x}_{2N}) & \dots & \varphi_{2N}(\mathbf{x}_{2N}) \end{pmatrix} \quad \#$$

and

$$\begin{aligned}
\langle \Phi | \Phi \rangle &= \\
& \int \det \begin{pmatrix} \varphi_1(\mathbf{x}_1) & \cdots & \varphi_{2N}(\mathbf{x}_1) \\ \vdots & \ddots & \vdots \\ \varphi_1(\mathbf{x}_{2N}) & \cdots & \varphi_{2N}(\mathbf{x}_{2N}) \end{pmatrix} \det \begin{pmatrix} \varphi_1(\mathbf{x}_1) & \cdots & \varphi_{2N}(\mathbf{x}_1) \\ \vdots & \ddots & \vdots \\ \varphi_1(\mathbf{x}_{2N}) & \cdots & \varphi_{2N}(\mathbf{x}_{2N}) \end{pmatrix} d\mathbf{x}_1 \cdots d\mathbf{x}_{2N} \\
&= \det \begin{pmatrix} \langle \varphi_1 | \varphi_1 \rangle & \cdots & \langle \varphi_1 | \varphi_{2N} \rangle \\ \vdots & \ddots & \vdots \\ \langle \varphi_{2N} | \varphi_1 \rangle & \cdots & \langle \varphi_{2N} | \varphi_{2N} \rangle \end{pmatrix}.
\end{aligned}$$

The invariance group of $|g_\rho^N\rangle$ is again $sp(2s, \mathbb{C})$ and the orbit of $|g_\rho^N\rangle$ under $GL(2s, \mathbb{C})$ produces a representation of the coset $\frac{GL(1, \mathbb{C}) \times SL(2s, \mathbb{C})}{sp(2s, \mathbb{C})}$ in $\mathcal{H}^{2N}$ by the manifold

$$|g_\rho(\tilde{\mathbf{a}}, \mathbf{z})^N\rangle = c_1(\tilde{\mathbf{a}})\tau_+(\mathbf{z})|g_\rho^N\rangle$$

which is parameterized by $\{\tilde{\mathbf{a}}, \mathbf{z}\}$. One can consider the manifold of normalized AGP's by restricting

$$\begin{aligned}
\tilde{\alpha}_{1r}(\tilde{\mathbf{a}}, \mathbf{z}) &= -\frac{1}{4N} \ln \{ \langle g_\rho(\tilde{\mathbf{a}}, \mathbf{z})^N | g_\rho(\tilde{\mathbf{a}}, \mathbf{z})^N \rangle \} \\
&= -\frac{1}{4N} \ln \{ \langle g_\rho^N | c(\tilde{\mathbf{a}})\tau_+(\mathbf{z})^\dagger \tau_+(\mathbf{z})c(\tilde{\mathbf{a}})g_\rho^N \rangle \} \\
&= -\frac{1}{4N} \ln \{ S_N(\tilde{\mathbf{a}}) \}.
\end{aligned}$$

The manifold generated by a reference AGP state of a given rank is an orbit of the group $GL(2s, \mathbb{C})$ and the group is transitive on that orbit (we can take ρ to label rank). The linear span of any such orbit is the whole space $\mathcal{H}^{2N}$ thus any reference AGP vector in $\mathcal{H}^{2N}$ is cyclic for $GL(2s, \mathbb{C})$.

Action of the group on GSO's

$$\gamma|g\rangle = \gamma \sum_{1 \leq j \leq s} c_j |\eta_{-j} \eta_j\rangle = \sum_{1 \leq j \leq s} c_j \{ |(\gamma \eta_{-j}) \eta_j\rangle + |\eta_{-j} (\gamma \eta_j)\rangle \}$$

$$\gamma|\eta_j\rangle = \gamma(\mathbf{z}, \boldsymbol{\eta}, \mathbf{w})|\eta_j\rangle = c(\boldsymbol{\eta}_{\text{re}})c(\boldsymbol{\eta}_{\text{im}})\tau_+(\mathbf{z})|\eta_j\rangle$$

$$\tau_+(\mathbf{z})|\eta_j\rangle = \exp \left\{ \sum_{\substack{-s \leq k < l \leq s \\ |k| \neq |l|}} z_{kl} q_{jl}^\dagger \right\} |\eta_j\rangle$$

$$= \exp \left\{ \sum_{\substack{-s \leq k < l \leq s \\ |k| \neq |l|}} a_p^\dagger a_q T_{pqkl} z_{kl} \right\} |\eta_j\rangle$$

Example of Indexing

The operators are indexed as

$$q_{jk} = (c_j a_j^\dagger a_k - \sigma(j)\sigma(k)c_k a_{-k}^\dagger a_{-j}); -s \leq j < k \leq s; -s \leq |j| \neq |k| \leq s \quad \#$$

with the symmetry

$$q_{jk} = (c_j a_j^\dagger a_k - \sigma(j)\sigma(k)c_k a_{-k}^\dagger a_{-j}) = \quad \#$$

$$- \sigma(jk)q_{-k-j} = -\sigma(jk)(c_k a_{-k}^\dagger a_{-j} - \sigma(j)\sigma(k)c_j a_j^\dagger a_k) \quad \#$$

and

$$q_{jk}^\dagger = (c_j a_k^\dagger a_j - \sigma(j)\sigma(k)c_k a_{-j}^\dagger a_{-k}); -s \leq j < k \leq s; -s \leq |j| \neq |k| \leq s. \quad \#$$

The diagonal blocks are given by

$$q_{j-j} = a_j^\dagger a_{-j} = q_{-jj}^\dagger \quad \#$$

$$q_{jj} = -q_{-j-j} = c_j(a_j^\dagger a_j - a_{-j}^\dagger a_{-j}) = q_{jj}^\dagger = -q_{-j-j}^\dagger. \quad \#$$

for $1 \leq j \leq s$.

$sp(4, \mathbb{C})$

The following example demonstrates the indexing for $sp(4, \mathbb{C})$. The basic pattern is

$$\begin{pmatrix} \# & -2 & -1 & 1 & 2 \\ -2 & \# & a & b & \# \\ -1 & c & \# & \# & b \\ 1 & d & \# & \# & a \\ 2 & \# & d & c & \# \end{pmatrix}. \quad \#$$

The off diagonal "block" elements are

$$q_{-22}, q_{2-2}, q_{-11}, q_{1-1}, \quad \#$$

where

$$q_{-22} = a_2^\dagger a_{-2}, q_{2-2} = a_{-2}^\dagger a_2 \quad \#$$

$$q_{-11} = a_1^\dagger a_{-1}, q_{1-1} = a_{-1}^\dagger a_1 \quad \#$$

and the diagonal ones are

$$q_{-2-2} = c_2(a_{-2}^\dagger a_{-2} - a_2^\dagger a_2) \quad \#$$

$$q_{-1-1} = c_1(a_{-1}^\dagger a_{-1} - a_1^\dagger a_1). \quad \#$$

The inter block basis elements are

$$q_{12} = c_1 a_1^\dagger a_2 - c_2 a_{-2}^\dagger a_{-1} = -q_{-2-1} \quad \#$$

$$q_{-12} = c_1 a_{-1}^\dagger a_2 + c_2 a_{-2}^\dagger a_1 = q_{-21} \quad \#$$

$$q_{1-2} = c_1 a_1^\dagger a_{-2} + c_2 a_2^\dagger a_{-1} = q_{2-1} \quad \#$$

$$q_{-1-2} = c_1 a_{-1}^\dagger a_{-2} - c_2 a_2^\dagger a_1 = -q_{21}. \quad \#$$

Thus the indexing is given by $\{q_{jk} | 1 \leq |j| < |k| \leq s - s \leq j, k \leq s\}$.

The commutation relationships based on

$$[q_{ij}, q_{kl}] = \sigma(jk) \{c_j c_k q_{il} \delta_{kj} + c_i c_l q_{-j-k} \delta_{il} + c_j c_l q_{i-k} \delta_{-jl} + c_i c_k q_{-j-l} \delta_{k-i}\}; \quad \#$$

$$-s \leq i < j \leq s; -s \leq k < l \leq s \quad \#$$

between inter block elements are given by

$$\begin{aligned}
[q_{12}, q_{-12}] &= [q_{-21}, q_{12}] = [(c_1 a_1^\dagger a_2 - c_2 a_{-2}^\dagger a_{-1}), (c_1 a_{-1}^\dagger a_2 + c_2 a_{-2}^\dagger a_1)] = & \# \\
&= -2c_1 c_2 a_{-2}^\dagger a_2 = 2c_1 c_2 q_{-22} & \# \\
[q_{12}, q_{1-2}] &= [(c_1 a_1^\dagger a_2 - c_2 a_{-2}^\dagger a_{-1}), (c_1 a_1^\dagger a_{-2} + c_2 a_2^\dagger a_{-1})] = & \# \\
&= 2c_1 c_2 a_1^\dagger a_{-1} = 2c_1 c_2 q_{1-1} & \# \\
[q_{12}, q_{-1-2}] &= -[q_{12}, q_{21}] = [(c_1 a_1^\dagger a_2 - c_2 a_{-2}^\dagger a_{-1}), (c_1 a_{-1}^\dagger a_{-2} - c_2 a_2^\dagger a_1)] = & \# \\
&= c_1 c_2 \{a_2^\dagger a_2 - a_{-2}^\dagger a_{-2} + a_{-1}^\dagger a_{-1} - a_1^\dagger a_1\} = c_1 c_2 \{q_{22} - q_{11}\} & \# \\
[q_{-21}, q_{1-2}] &= [(c_2 a_{-2}^\dagger a_1 + c_1 a_{-1}^\dagger a_2), (c_1 a_1^\dagger a_{-2} + c_2 a_2^\dagger a_{-1})] = & \# \\
&= c_1 c_2 a_{-2}^\dagger a_{-2} - c_1 c_2 a_{-1}^\dagger a_{-1} + c_1 c_2 a_{-1}^\dagger a_{-1} - c_1 c_2 a_{-2}^\dagger a_{-2} = 0 & \# \\
[q_{-21}, q_{-1-2}] &= [(c_2 a_{-2}^\dagger a_1 + c_1 a_{-1}^\dagger a_2), (c_1 a_{-1}^\dagger a_{-2} - c_2 a_2^\dagger a_1)] = & \# \\
&= c_2 c_1 a_{-1}^\dagger a_1 - c_2 c_1 a_{-1}^\dagger a_1 = 0 & \# \\
[q_{1-2}, q_{-1-2}] &= [q_{21}, q_{1-2}] = [(c_1 a_1^\dagger a_{-2} + c_2 a_2^\dagger a_{-1}), (c_1 a_{-1}^\dagger a_{-2} - c_2 a_2^\dagger a_1)] = & \# \\
&= 2c_1 c_2 a_2^\dagger a_{-2} = 2c_1 c_2 q_{2-2} & \# \\
[q_{12}, a_2^\dagger a_{-2}] &= [q_{12}, q_{2-2}] = [(c_1 a_1^\dagger a_2 - c_2 a_{-2}^\dagger a_{-1}), a_2^\dagger a_{-2}] = & \# \\
&= c_1 a_1^\dagger a_{-2} + c_2 a_2^\dagger a_{-1} = q_{1-2} & \# \\
[q_{12}, a_1^\dagger a_{-1}] &= [q_{12}, q_{1-1}] = [(c_1 a_1^\dagger a_2 - c_2 a_{-2}^\dagger a_{-1}), a_1^\dagger a_{-1}] = 0 & \# \\
[q_{12}, (a_1^\dagger a_1 - a_{-1}^\dagger a_{-1})] &= [(c_1 a_1^\dagger a_2 - c_2 a_{-2}^\dagger a_{-1}), (a_1^\dagger a_1 - a_{-1}^\dagger a_{-1})] = 0 & \# \\
[q_{12}, (a_2^\dagger a_2 - a_{-2}^\dagger a_{-2})] &= [q_{12}, q_{22}] = [(c_1 a_1^\dagger a_2 - c_2 a_{-2}^\dagger a_{-1}), (a_2^\dagger a_2 - a_{-2}^\dagger a_{-2})] & \# \\
&= c_1 a_1^\dagger a_2 - c_2 a_{-2}^\dagger a_{-1} = q_{12} & \# \\
[a_1^\dagger a_{-1}, (a_1^\dagger a_1 - a_{-1}^\dagger a_{-1})] &= -2a_1^\dagger a_{-1} & \# \\
[a_2^\dagger a_{-2}, (a_2^\dagger a_2 - a_{-2}^\dagger a_{-2})] &= -2a_2^\dagger a_{-2} & \#
\end{aligned}$$

$sp(6, \mathbb{C})$

The following example demonstrates the indexing for $sp(6, \mathbb{C})$

	-3	-2	-1	1	2	3	
-3	#	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	#	
-2	<i>e</i>	#	<i>f</i>	<i>g</i>	#	<i>d</i>	
-1	<i>h</i>	<i>i</i>	#	#	<i>g</i>	<i>c</i>	#
1	<i>j</i>	<i>k</i>	#	#	<i>f</i>	<i>b</i>	
2	<i>l</i>	#	<i>k</i>	<i>i</i>	#	<i>a</i>	
3	#	<i>l</i>	<i>j</i>	<i>h</i>	<i>e</i>	#	

The "block diagonal" operators are

$$q_{-3-3}, q_{-2-2}, q_{-1-1}; \quad \#$$

where

$$q_{-3-3} = c_3(a_{-3}^\dagger a_{-3} - a_3^\dagger a_3) \quad \#$$

$$q_{-2-2} = c_2(a_{-2}^\dagger a_{-2} - a_2^\dagger a_2) \quad \#$$

$$q_{-1-1} = c_1(a_{-1}^\dagger a_{-1} - a_1^\dagger a_1) \quad \#$$

and the off diagonal operators are

$$q_{-33}, q_{3-3}, q_{-22}, q_{2-2}, q_{-11}, q_{1-1}, \quad \#$$

where

$$q_{-33} = a_3^\dagger a_{-3}; q_{3-3} = a_{-3}^\dagger a_3 \quad \#$$

$$q_{-22} = a_2^\dagger a_{-2}; q_{2-2} = a_{-2}^\dagger a_2 \quad \#$$

$$q_{-11} = a_1^\dagger a_{-1}; q_{1-1} = a_{-1}^\dagger a_1. \quad \#$$

$$[q_{ij}, q_{kl}] = \sigma(jk) \{c_i q_{il} \delta_{kj} + c_i c_l q_{-j-k} \delta_{il} + c_i c_l q_{i-k} \delta_{-jl} + c_i c_l q_{-jl} \delta_{k-i}\}; \quad \#$$

$$-s \leq i < j \leq s; -s \leq k < l \leq s \quad \#$$

The commutation relationships between the block operators are

$$[q_{-3-3}, q_{-33}] = -[c_3(a_3^\dagger a_3 - a_{-3}^\dagger a_{-3}), a_3^\dagger a_{-3}] = -c_3(a_3^\dagger a_3 a_3^\dagger a_{-3} + a_3^\dagger a_{-3} a_3^\dagger a_3) \quad \#$$

$$= -c_3 a_3^\dagger a_{-3} = c_3 q_{-33}, \quad \#$$

$$[q_{2-3}, q_{-3-3}] = \quad \#$$

$$- [c_3(a_3^\dagger a_3 - a_{-3}^\dagger a_{-3}), (c_3 a_{-3}^\dagger a_2 - \sigma(-3)\sigma(2)c_2 a_{-2}^\dagger a_3)] \quad \#$$

$$= c_3^2 a_{-2}^\dagger a_3 - c_2 c_3 a_{-3}^\dagger a_2 = c_3 q_{2-3}. \quad \#$$

The off block operators are

$$q_{12}, q_{1-2}, q_{13}, q_{1-3}, q_{-12}, q_{-1-2}, q_{-13}, q_{-1-3}, q_{23}, q_{2-3}, q_{-23}, q_{-2-3} \quad \#$$

are indexed by

$$\{q_{jk} | 1 \leq |j| < |k| \leq s - s \leq j, k \leq s\}, \quad \#$$

where

$$\begin{aligned}
q_{12} &= c_1 a_1^\dagger a_2 - c_2 a_{-2}^\dagger a_{-1} & \# \\
q_{1-2} &= c_1 a_1^\dagger a_{-2} + c_2 a_2^\dagger a_{-1} & \# \\
q_{-12} &= c_1 a_{-1}^\dagger a_2 + c_2 a_{-2}^\dagger a_1 & \# \\
q_{-1-2} &= c_1 a_{-1}^\dagger a_{-2} - c_2 a_2^\dagger a_1 & \# \\
q_{13} &= c_1 a_1^\dagger a_3 - c_3 a_{-3}^\dagger a_{-1} & \# \\
q_{1-3} &= c_1 a_1^\dagger a_{-3} + c_3 a_3^\dagger a_{-1} & \# \\
q_{-13} &= c_1 a_{-1}^\dagger a_3 + c_3 a_{-3}^\dagger a_1 & \# \\
q_{-1-3} &= c_1 a_{-1}^\dagger a_{-3} - c_3 a_3^\dagger a_1 & \# \\
q_{23} &= c_2 a_2^\dagger a_3 - c_3 a_3^\dagger a_2 & \# \\
q_{2-3} &= c_2 a_2^\dagger a_{-3} + c_3 a_3^\dagger a_{-2} & \# \\
q_{-23} &= c_2 a_{-2}^\dagger a_3 + c_3 a_{-3}^\dagger a_2 & \# \\
q_{-2-3} &= c_2 a_{-2}^\dagger a_{-3} - c_3 a_3^\dagger a_2 & \#
\end{aligned}$$

The commutation relationships are based on

$$[a_r^\dagger a_s, a_p^\dagger a_q] = a_r^\dagger a_q \delta_{sp} - a_p^\dagger a_s \delta_{rq} \quad \#$$

and are given by

$$\begin{aligned}
[q_{12}, q_{13}] &= [c_1 a_1^\dagger a_2 - c_2 a_{-2}^\dagger a_{-1}, c_1 a_1^\dagger a_3 - c_3 a_{-3}^\dagger a_{-1}] = 0 & \# \\
[q_{12}, q_{1-3}] &= [c_1 a_1^\dagger a_2 - c_2 a_{-2}^\dagger a_{-1}, c_1 a_1^\dagger a_{-3} + c_3 a_3^\dagger a_{-1}] = 0 & \# \\
[q_{12}, q_{-13}] &= [c_1 a_1^\dagger a_2 - c_2 a_{-2}^\dagger a_{-1}, c_1 a_{-1}^\dagger a_3 - c_3 a_{-3}^\dagger a_1] = & \# \\
& c_1 c_3 a_{-3}^\dagger a_2 - c_2 c_1 a_{-2}^\dagger a_3 = c_1 (c_3 a_{-3}^\dagger a_2 - c_2 a_{-2}^\dagger a_3) = c_1 q_{-32} & \# \\
[q_{12}, q_{-1-3}] &= [c_1 a_1^\dagger a_2 - c_2 a_{-2}^\dagger a_{-1}, c_1 a_{-1}^\dagger a_{-3} - c_3 a_3^\dagger a_{-1}] = & \# \\
& c_1 c_3 a_3^\dagger a_2 - c_1 c_2 a_{-2}^\dagger a_{-3} = c_1 (c_3 a_3^\dagger a_2 - c_2 a_{-2}^\dagger a_{-3}) = c_1 q_{32} & \# \\
[q_{12}, q_{23}] &= [c_1 a_1^\dagger a_2 - c_2 a_{-2}^\dagger a_{-1}, c_2 a_2^\dagger a_3 - c_3 a_{-3}^\dagger a_{-2}] = & \# \\
& c_2 (c_1 a_1^\dagger a_3 - c_3 a_{-3}^\dagger a_{-1}) = c_2 q_{13} & \# \\
[q_{12}, q_{2-3}] &= [c_1 a_1^\dagger a_2 - c_2 a_{-2}^\dagger a_{-1}, c_2 a_2^\dagger a_{-3} - c_3 a_3^\dagger a_{-2}] = & \# \\
& -c_2 (c_1 a_1^\dagger a_{-3} + c_3 a_3^\dagger a_{-1}) = -c_2 q_{1-3} & \# \\
[q_{12}, q_{-23}] &= [c_1 a_1^\dagger a_2 - c_2 a_{-2}^\dagger a_{-1}, c_2 a_{-2}^\dagger a_3 - c_3 a_{-3}^\dagger a_2] = 0 & \# \\
[q_{12}, q_{-2-3}] &= [c_1 a_1^\dagger a_2 - c_2 a_{-2}^\dagger a_{-1}, c_2 a_{-2}^\dagger a_{-3} - c_3 a_3^\dagger a_2] = 0 & \#
\end{aligned}$$

which covers and corresponds to $-3 \leq j \leq k \leq 3$.

Lie Algebra Generators

A reducible representation $R(gl(2s, \mathbb{C}))$ of the Lie algebra $gl(2s, \mathbb{C})$ on $\mathcal{H}_F$ is generated by

$$\begin{aligned}
a_{jk}^\dagger &= c_j a_k^\dagger a_j - \sigma(j)\sigma(k)c_k a_{-j}^\dagger a_{-k}; 1 \leq j < k \leq s; |j| \neq |k| & \# \\
a_{jk} &= c_j a_j^\dagger a_k - \sigma(j)\sigma(k)c_k a_{-k}^\dagger a_{-j}; 1 \leq j < k \leq s; |j| \neq |k| & \# \\
a_j^\dagger a_k; &|j| = |k|; 1 \leq j \leq s. & \#
\end{aligned}$$

In line with Gilmore (where $q \equiv Z$) one defines

$$q_{j-j} = a_j^\dagger a_{-j} \quad \#$$

$$q_{-jj} = a_{-j}^\dagger a_j = q_{j-j}^\dagger. \quad \#$$

The number of q generators these generators is

$$2s(2s-1) - 2 = 4s^2 - 4s \quad \#$$

and the number of subblock generators is

$$4s \quad \#$$

giving a total number

$$4s^2, \quad \#$$

which equals the number of generators $\{a_j^\dagger a_k \mid -s \leq j, k \leq s\}$. The representation is the sum of the irreps

$$R(gl(2s, \mathbb{C})) = \bigoplus_{0 \leq N \leq 2s} R_N(gl(2s, \mathbb{C})), \quad \#$$

where $\{R_N(gl(2s, \mathbb{C})) \mid 0 \leq N \leq r\}$ are the irreducible representations carried by $\mathcal{H}^N$.

The $\{q_{jk}, q_{jk}^\dagger \mid 1 \leq j < k \leq s; |j| \neq |k|\}$ generators can be expressed in terms of $\{a_j^\dagger a_k \mid -s \leq j < k \leq s; |j| \neq |k|\}$, when $c_j \neq c_k$, as

$$\begin{pmatrix} q_{-j-k}^\dagger \\ q_{j-k}^\dagger \\ q_{-jk}^\dagger \\ q_{jk}^\dagger \\ q_{-j-k} \\ q_{j-k} \\ q_{-jk} \\ q_{jk} \end{pmatrix} = \begin{pmatrix} c_k & 0 & \cdots & \cdots & \cdots & \cdots & 0 & -c_j \\ 0 & c_k & 0 & \cdots & \cdots & 0 & c_j & 0 \\ \vdots & 0 & c_k & 0 & 0 & c_j & 0 & \vdots \\ \vdots & \vdots & 0 & c_k & -c_j & 0 & \vdots & \vdots \\ \vdots & \vdots & 0 & -c_j & c_k & 0 & \vdots & \vdots \\ \vdots & 0 & c_j & 0 & 0 & c_k & 0 & \vdots \\ 0 & c_j & 0 & \cdots & \cdots & 0 & c_k & 0 \\ -c_j & 0 & \cdots & \cdots & \cdots & \cdots & 0 & c_k \end{pmatrix} \begin{pmatrix} a_{-j}^\dagger a_{-k} \\ a_{-j}^\dagger a_k \\ a_j^\dagger a_{-k} \\ a_j^\dagger a_k \\ a_{-k}^\dagger a_{-j} \\ a_k^\dagger a_{-j} \\ a_{-k}^\dagger a_j \\ a_k^\dagger a_j \end{pmatrix}. \quad \#$$

$$\begin{pmatrix} u_{jk} \\ u_{j+sk} \\ u_{jk+s} \\ u_{j+sk+s} \\ v_{jk} \\ v_{j+sk} \\ v_{jk+s} \\ v_{j+sk+s} \end{pmatrix} = \begin{pmatrix} c_k & 0 & \cdots & \cdots & \cdots & \cdots & 0 & c_j \\ 0 & c_k & 0 & \cdots & \cdots & 0 & -c_j & 0 \\ \vdots & 0 & c_k & 0 & 0 & -c_j & 0 & \vdots \\ \vdots & \vdots & 0 & c_k & c_j & 0 & \vdots & \vdots \\ \vdots & \vdots & 0 & -c_j & c_k & 0 & \vdots & \vdots \\ \vdots & 0 & c_j & 0 & 0 & c_k & 0 & \vdots \\ 0 & c_j & 0 & \cdots & \cdots & 0 & c_k & 0 \\ -c_j & 0 & \cdots & \cdots & \cdots & \cdots & 0 & c_k \end{pmatrix} \begin{pmatrix} a_j^\dagger a_k \\ a_j^\dagger a_{k+s} \\ a_{j+s}^\dagger a_k \\ a_{j+s}^\dagger a_{k+s} \\ a_k^\dagger a_j \\ a_k^\dagger a_{j+s} \\ a_{k+s}^\dagger a_j \\ a_{k+s}^\dagger a_{j+s} \end{pmatrix}, \quad \#$$

$$\begin{aligned}
[\mathcal{X}_{jk}, \mathcal{X}_{lm}] &= \left[\sum_{\substack{-s \leq j, k \leq s, |j| \neq |k| \\ 1 \leq \alpha, \beta \leq 8}} c_{jka\beta} a_{\alpha}^{\dagger} a_{\beta}, \sum_{\substack{-s \leq l, m \leq s, |l| \neq |m| \\ 1 \leq \gamma, \delta \leq 8}} c_{lm\gamma\delta} a_{\gamma}^{\dagger} a_{\delta} \right] \quad \# \\
&= \sum_{\substack{-s \leq j, k \leq s, |j| \neq |k| \\ 1 \leq \alpha, \beta \leq 8 \\ -s \leq l, m \leq s, |l| \neq |m| \\ 1 \leq \gamma, \delta \leq 8}} c_{jka\beta} c_{lm\gamma\delta} [a_{\alpha}^{\dagger} a_{\beta}, a_{\gamma}^{\dagger} a_{\delta}] \quad \#
\end{aligned}$$

Commutation Relationships

The commutation relations for $sp(2s, \mathbb{C})$ are

$$\begin{aligned}
[q_{ij}, q_{kl}] &= \sigma(jk) \{q_{il} \delta_{kj} + q_{-j-k} \delta_{il} + q_{i-k} \delta_{-jl} + q_{-jl} \delta_{k-i}\}; \quad \# \\
-s \leq i < j \leq s; -s \leq k < l \leq s &\quad \#
\end{aligned}$$

and the symmetry relationships

$$q_{ij} = -\sigma(ij) q_{-j-i} \quad \#$$

are satisfied.

$$\begin{aligned}
[a_r^{\dagger} a_s, a_p^{\dagger} a_q] &= a_r^{\dagger} a_q \delta_{sp} - a_p^{\dagger} a_s \delta_{rq} \quad \# \\
[a_r^{\dagger} a_s, a_p^{\dagger} a_p] &= a_r^{\dagger} a_p \delta_{sp} - a_p^{\dagger} a_s \delta_{rp} \quad \# \\
&= \begin{cases} a_r^{\dagger} a_p, s = p, r \neq s \\ -a_p^{\dagger} a_s, r = p, r \neq s \end{cases} \quad \# \\
[(c_j a_k^{\dagger} a_j - \sigma(j) \sigma(k) c_k a_{-j}^{\dagger} a_{-k}), a_l^{\dagger} a_m] &= \quad \# \\
c_j (a_k^{\dagger} a_m \delta_{jl} - a_l^{\dagger} a_j \delta_{km}) - c_k \sigma(j) \sigma(k) (a_{-j}^{\dagger} a_m \delta_{-kl} - a_l^{\dagger} a_{-k} \delta_{-jm}) &\quad \# \\
m = l = j : [,] = c_j a_k^{\dagger} a_m - c_k \sigma(j) \sigma(k) &\quad \#
\end{aligned}$$

If one shifts the indexing by

$$\begin{aligned}
v(j) &= 2j, 1 \leq j \leq s \quad \# \\
v(j) &= -(2j+1), -s \leq j \leq -1 \quad \#
\end{aligned}$$

then

$$[q_{ij}, q_{jk}] = q_{ik}; 1 \leq i < j < k \leq 2s \quad \#$$

and

$$[q_{jk}^{\dagger}, q_{ij}^{\dagger}] = -[q_{ij}^{\dagger}, q_{jk}^{\dagger}] = q_{ik}^{\dagger}; 1 \leq i < j < k \leq 2s. \quad \#$$

The general linear group $GL(\mathcal{H}^1)$ can be factored into the product

$$GL(\mathcal{H}^1) = GL(1) \times SL(\mathcal{H}^1), \quad \#$$

such that

$$g = (\det g) h = e^{Tr \langle g \rangle} h, h \in SL(\mathcal{H}). \quad \#$$

where

$$\eta_j = Tr\{\lambda_j\}$$

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